

Jaya Sharma

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INTRODUCTION

I'm a PhD student in Cybersecurity Engineering at the Technical University of Denmark (DTU), working under the supervision of Dr. Christian Majenz. My current research lies at the intersection of theoretical cryptography and quantum computing, with a focus on the security of cryptographic constructions in the quantum random oracle model. My other specific interests include public key cryptography, zero-knowledge proofs, lattice-based cryptography and quantum information.

EDUCATION

Technical University of Denmark (DTU) PhD in Cybersecurity Engineering, <i>Supervisor:</i> Dr. Christian Majenz	2024-27
Indraprastha Institute of Information Technology Delhi (IIITD) PhD Junior Research Fellow in Computer Science and Engineering (<i>completed 3 semesters</i>)	2022-24
Indian Institute of Technology Roorkee (IITR) Master of Science in Mathematics	2020-22 Grade: 8.675/10
University of Delhi- Lady Shri Ram College For Women (LSR) Bachelor of Science (Honors) in Mathematics	2017-20 Grade: 9.081/10

RESEARCH VISITS/INTERNSHIPS

Centrum Wiskunde & Informatica (CWI) <i>Host:</i> Dr. Serge Fehr -Working on post-quantum non interactive zero knowledge proofs.	09/2026-01/2027
QWorld Association-QIntern (<i>Remote internship</i>) <i>Project:</i> Implementing Efficient block-encoding of dense matrices via QRAM.	07-09/2025
Indian Institute of Science, Bangalore (IISc) <i>Host:</i> Dr. Chaya Ganesh -Studied compression theory of sigma protocols and lattice based sigma protocols.	06-07/2023
Indian Institute of Science Education and Research (IISER) Bhopal (<i>Remote internship</i>) <i>Host:</i> Dr. Sanjay Singh <i>Project:</i> Introduction to Algebraic Function Fields and Codes	06-08/2021

PUBLICATIONS/PROJECTS

Security of the Fischlin's Transform in Quantum Random Oracle Model <i>QCrypt'26, IACR ASIACRYPT'26</i> Christian Majenz, Jaya Sharma	2026
Implementating Efficient block-encoding of dense matrices via QRAM using the Qrisp framework <i>QWorld QIntern Project 2025 (Library currently in review by Qrisp Developers)</i> Viraj Dsouza, Kalpesh Bhatnagar, Kyle Carlo Lasala, Jaya Sharma	2025
A Survey on Quantum Error Correcting Codes	2023
Elliptic Curves Over Complex Numbers- Master's Thesis Supervisor- Dr. A. Swaminathan	2022
Brent-Dekker Method: An Attempt to Decode <i>Eclat, A Mathematics Journal, Vol 11</i> Dr. Monika Singh, Shagun Agrawal, Jaya Sharma	2020

PROGRAMMING

C++, [Python](#), Qiskit.

PRESENTATIONS

Security of the Fischlin Transform in Quantum Random Oracle Model- QCrypt	08/2026
Fiat Shamir Transform in QROM- Crypto Reading Group DTU	10/2024
Lattice Based Sigma Protocols- Theory Reading Group IIITD	10/2023
Trapdoors for Hard Lattices and New Cryptographic Constructions-Theory Reading Group IIITD	05/2023
The Rabin Cryptosystem: Mathematical Foundations and Implementation- IIITD	12/2022
Coherent Measures of Financial Risk- MSc Mathematics Seminar, IITR	10/2021
Elliptic Curve Cryptography- Department of Mathematics, LSR	05/2019
Latin Squares- Recreation, Disproving Euler to Cryptography- Department of Mathematics, LSR	05/2018
Applications of Congruence Modulo- Department of Mathematics LSR	02/2018

TEACHING ASSISTANCE

Discrete Mathematics (DTU)	2024,2025
Graph Theory (DTU)	2025
Applied Cryptography (DTU)	2024
Introduction To Cryptography (IIT Delhi)	2023
Discrete Mathematics (IIITD)	2022

AWARDS

Thomas B. Thriges Fond Grant, William Demant Fonden Grant, Otto Mønstedts Fond Grant -Danish Grants to fund research stay at CWI Amsterdam.	2026
ACM-India Anveshan Setu Fellowship - Awarded by Association for Computing Machinery(ACM) India to work at IISc.	2023
Joint CSIR-UGC NET (JRF) Fellowship Mathematical Sciences -Awarded PhD Research Fellowship for securing All India Rank 122/30942 candidates in the exam conducted jointly by Council for Scientific and Industrial Research and University Grants Commission India.	2022
IIT JAM Mathematics: secured All India Rank 143/14374 candidates - Joint Admission Test for Master of Science Courses at Centrally Funded Technical Institutions in India.	2020

WORKSHOPS/SUMMER SCHOOLS/CONFERENCES

Qcrypt- Ottawa, Canada	08/2026
Eurocrypt- Rome, Italy	05/2026
DQC Scientific Quantum Conference 2026- Aalborg University, Denmark	04/2026
Theory of Cryptography Conference (TCC) 2025-Aarhus, Denmark	12/2025
Workshop on Mathematics of Post-Quantum Cryptography- University of Zurich	06/2025
Quantum Safe Internet (QSI) Workshop- Technical University of Denmark, Lyngby	05/2025
IACR Summer School on Post Quantum Cryptography- University of Warsaw	07/2024
ACM India Grad Cohort- IISc Bangalore	07/2023
CSE Research Symposium- IIT Bombay	03/2023
Vigyan Vidushi Mathematics Summer School- TIFR Mumbai	07/2021
STCS Vigyan Vidushi Summer School on Theoretical Computer Science- TIFR Mumbai	06/2021
Workshop: Calculus the Geometrical Way- Miranda House College, University of Delhi	09/2019

OTHER ACTIVITIES

PhD Committee Member- DTU Compute	2025
Illustrator-The LSR College Magazine	2018-19

LANGUAGES

English, Danish (elementary, completed DU3 A2 level), Hindi (mother tongue), Sanskrit (elementary proficiency).